NOVOGENIA · ACADEMY

# FREQUENTLY ASKED QUESTIONS BIOLOGICAL AGE

Answers to typical client questions about the biological age analysis.

CONTENTS


01 BASICS — TELOMERES & AGING

02 WHAT LIFESTYLE DOES

03 INFLUENCE & EXPECTATIONS


## 01 · BASICS — TELOMERES & AGING

### What are telomeres and why are they important for aging?

Telomeres are protective caps at the end of each chromosome — they contain no genes themselves, but protect genes from being lost during cell divisions. With each cell division, a small piece of telomere breaks off. After 50-70 divisions, they are used up; the cell goes into a sleep mode (senescence) and accumulates as an "old cell" in the body.

### What is the difference between chronological and biological age?

Chronological age is the time you have lived, in years — fixed and unchangeable. Biological age describes how “old” your cells actually are, and telomeres are one of the factors behind it: babies have long telomeres, 90-year-olds short ones. People with two defective copies of the telomerase gene are already born with shorter telomeres — the difference in biological age can be up to 20 years.

### What does telomerase do?

The telomerase gene helps maintain the telomeres. It cannot completely stop or reverse their shortening, but it slows it down: with well-functioning genes, telomeres get shorter more slowly with age; with a gene defect, they are lost faster. How well this protection works is partly genetic — and that is what the analysis looks at.

### Which genes are evaluated in the analysis?

The analysis looks at two things: first, your genetic ability to maintain your telomeres, which depends mainly on the telomerase gene; second, whether a Mediterranean diet has an effect on your telomeres, which also depends on certain gene variants. It shows your genetic predisposition — not how long your telomeres actually are.

## 02 · WHAT LIFESTYLE DOES

### Does a Mediterranean diet help maintain telomeres in everyone?

Surprisingly, no. Studies have shown that a Mediterranean diet helps keep telomeres longer only in people with certain gene variants — in everyone else, it has no effect on the telomeres.

Your report shows which group you belong to. This refers only to the telomeres, not to other effects of the diet — and it is a clear example that the same lifestyle advice does not work the same way for everyone.

### Which nutrients support telomeres?

Vitamin D3, Omega-3, Vitamin E, Resveratrol and Ginkgo. When the genetic protection system works less well, increasing these nutrients — for example by eating more foods that contain them — can help slow telomere shortening. They cannot reverse it.

### Can I influence free radicals?

Yes, to a degree. Free radicals continuously damage our cells and are one of the reasons we age. Antioxidants help counter this; they are found, for example, in foods rich in Vitamin C and Vitamin E or in polyphenols from berries and tea. An antioxidant-rich diet can therefore support healthy aging.

## 03 · INFLUENCE & EXPECTATIONS

### How quickly will I see results from anti-aging measures?

Not overnight. The recommended measures cannot make telomeres grow back — they help slow down their shortening, so the benefit builds up gradually over many years. Note that the genetic analysis shows your predisposition, not the current length of your telomeres, so the analysis result itself does not change through these measures.

### What practical use does this knowledge give me?

You know how well your genes maintain your telomeres — and whether a Mediterranean diet helps you in this respect. If telomere maintenance is impaired, you can respond deliberately by increasing Vitamin D3, Omega-3, Vitamin E, Resveratrol and Ginkgo, for example through foods that contain them. A generally healthy lifestyle with an antioxidant-rich diet, regular exercise and enough sleep is also sensible.